d- Block elements composed of 3d series (4th period) Scandium to Zinc (10 elements), 4d series (5th period) Yttrium to Cadmium (10 elements) and 5d series (6th period) Lanthanum, Haffinium to mercury. As we know that the group-12 elements Zinc, Cadmium and Mercury do not have partially filled d-orbital either in their elemental state or in their normal oxidation states. However they are treated as transition elements, because their properties are an extension of the properties of the respective transition elements. As per the IUPAC definition, the seventh period elements, starting from Ac, Rf to Cn also belong to transition metals. All of them are radioactive. Except Actinium; all the remaining elements are synthetically prepared and have very low half life periods.

4.2 Electronic configuration:

We have already learnt in XI STD to write the electronic configuration of the elements using Aufbau principle, Hund's rule etc. According to Aufbau principle, the electron first fills the 4s orbital before 3d orbital. Therefore filling of 3d orbital starts from Sc, its electronic configuration is [Ar]$3d^1$ $4s^2$ and the electrons of successive elements are progressively filled in 3d orbital and the filling of 3d orbital is complete in Zinc, whose electronic configuration is [Ar] $3d^{10}$ $4s^2$. However, there are two exceptions in the above mentioned progressive filling of 3d orbitals; if there is a chance of acquiring half filled or fully filled 3d sub shell, it is given priority as they are the stable configuration, for example Cr and Cu.

The electronic configurations of Cr and Cu are [Ar] $3d^5$ $4s^1$ and [Ar] $3d^{10}$ $4s^1$ respectively. The extra stability of half filled and fully filled d orbitals, as already explained in XI STD, is due to symmetrical distribution of electrons and exchange energy.

Note: The extra stability due to symmetrical distribution can also be visualized as follows. When the d orbitals are considered together, they will constitute a sphere. So the half filled and fully filled configuration leads to complete symmetrical distribution of electron density. On the other hand, an unsymmetrical distribution of electron density as in the case of partially filled configuration will result in building up of a potential difference. To decrease this and to achieve a tension free state with lower energy, a symmetrical distribution is preferred.

With these two exceptions and minor variation in certain individual cases, the general electronic configuration of d- block elements can be written as [Noble gas] $(n-1)d^{1-10}ns^{1-2}$, Here, $n = 4$ to 7. In periods 6 and 7, (except La and Ac) the configuration includes $((n-2)f$ orbital; [Noble gas] $(n-2)f^{14}(n-1)d^{1-10}ns^{1-2}$.

4.3 General trend in properties:

4.3.1 Metallic behaviour:

All the transition elements are metals. Similar to all metals the transition metals are good conductors of heat and electricity. Unlike the metals of Group-1 and group-2, all the transition metals except group 11 elements are hard. Of all the known elements, silver has the highest electrical conductivity at room temperature.

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